

Question number	Answer	Additional guidance	Mark
7(a)(i)	{the number of / all} humans in the world (1)	ACCEPT amount people / <i>Homo sapiens</i> Earth / this planet / globally	(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	producing enough <u>food</u> / producing <u>food</u> without damaging the environment (1)	ACCEPT without running out minimises carbon footprint so food can be made in the future	(1)

Question number	Answer	Additional guidance	Mark
7(b)(i)	An explanation that includes the following points: • reason for release of carbon dioxide into the atmosphere by land clearance (1) • reason for release of carbon dioxide into the atmosphere by farming (1) • reason for less carbon dioxide being removed from the atmosphere (1) • reason for release of methane into the atmosphere (1) • credit details of how green house gases cause global warming (1)	e.g. burning of forests, burning of (fossil) fuels by vehicles, decomposition of dead remains by bacteria e.g. burning of (fossil) fuels by {tractors, transport of food products, factories} IGNORE by respiration e.g. trees absorbed more carbon dioxide than crops, animals do not remove carbon dioxide from the atmosphere e.g. paddy fields, cattle, decomposition of dead remains by bacteria e.g. trap heat energy	(3)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	71% of 149 million calculated (1) - farmed area calculated (1) $5.29 \times 10^7 / 5.3 \times 10^7 (\text{km}^2)$ (1)	e.g. of calculation 105 790 000 $(105\,790\,000 \div 2 =) \, 52\,895\,000$ ecf incorrect answer or 105 790 000 expressed in standard form to 1 or 2 dps correctly Bald answer of $5.29 \times 10^7 / 5.3 \times 10^7 (\text{km}^2) = 3$ marks Bald answer of 52 895 000 or incorrect standard form for this value = 2 marks Bald answer of $1.06 \times 10^8 = 2$ marks Bald answer of 105 790 000 = 1 mark	(3)

Question number	Answer	Additional guidance	Mark
7(b)(iii)	values in range of 1 : 0.06 to 1 : 0.1 (with a max of 2 decimal places) or 1 : 11 to 1 : 15 (1)	ACCEPT ratio expressed either way around	(1)

Question number	Answer	Additional guidance	Mark
*7(c)	- recycling of plant material (D) - to add nutrients to ground - so that artificial fertiliser use is reduced - which will cause less harm to the environment - genetic engineering to produce more {insect / drought} resistant crops (D) - to grow higher yield crops - so that more food is produced from the land - and fewer habitats have to be destroyed - reduce air miles to transport food (D) - use biofuels to fuel machinery used in farming (D) - use waste crop materials to make biofuels (D) - so that less carbon dioxide is released into the air - raise fewer animals and grow more crops (D) - because animals release carbon dioxide into atmosphere - which contribute to greenhouse effect - plants absorb more carbon dioxide from atmosphere - use solar / wind power for energy need in farming (D) - as these are sustainable - and do not produce carbon dioxide - encourage population to eat more plant-based foods (D) - as more mass of food can be produced from the land - because fewer trophic levels therefore less energy lost from the food chain - credit examples given from the diagram - reduce the quantity of plastic in packaging of food (D) - because it is not biodegradable - because it can harm animals	Level 1: 1 mark = one aspect described 2 marks = one aspect described with a simple explanation Level 2: 3 marks = one aspect described with an extended explanation OR two aspects described each with a simple explanation 4 marks = two aspects described each with an extended explanation OR three aspects described each with a simple explanation Level 3: 5 marks = three aspects described, all with extended explanation 6 marks = three aspects described, all with extended explanation that includes energy loss from a food chain	(6)